Assignment 1 - Measures of Central Tendency

Task : Load a small dataset of numbers (e.g., exam scores). Compute mean, median and mode

~~Hint: Use mean~~

data set ÷ exam_scores = ([85, 90, 88, 92, ~~80~~ 85, 78, 90, 95, 88, 90])

libraries used = numpy , statistics

numpy.mean (exam_scores) // Use to find the average

numpy.~~mode~~ median (exam_scores) // Use to find the middle value

statistics.mode (exam_scores) // Use to find the value which occurs the most in the data set

Here, we are using statistics library because numpy doesn't have 'mode' function.

Assignment 2 - Measures of Variation

Task : Using the same dataset, compute range, variance, standard deviation

Hint : var(), std(), max() - min()

Goal : See how spread affects interpretation (stable vs volatile data)

Libraries used : numpy.

A bit of code :

```
data_range = max(data) - min(data)
data_variance = np.var(data, ddof=1) // The ddof
data_std = np.std(data, ddof=1)
```

ddof stands for Delta degrees of freedom which control denominator when calculating variance and standard variation.

// ddof = 0 → uses denominator n
→ population variance

// ddof = 1 → uses denominator n-1
→ sample variance.

// population variation is used when we have all the data otherwise we use sample ~~formula~~ variance.

Assignment 3 - Summarizing Data with Pandas

Task : Load a CSV dataset. Use describe() to generate summary statistics.

Goal : Learn to quickly summarize datasets in EDA phase.

Code :

```
import pandas as pd
df = pd.read_csv("file.csv")
print(df.head()) // print first few columns of the dataset
summary = df.describe()
print(summary)
```

// .describe() shows :

- count → no. of entries
- mean
- std
- min
- 25%, 50%, 75% → Outliers
- max

Assignment 4 - Outlier Detection using IQR

Task : Identify outliers in a dataset using Interquartile Range (IQR)

Goal : Connect measures of spread with anomaly detection in Data Science.

Libraries used : Pandas.

Snippets of code :

```
Q1 = df["Math"].quantile(0.25)  // (first quartile, 25%)
Q3 = df["Math"].quantile(0.75)  // (third quartile, 75%)
IQR = Q3 - Q1   // (middle 50% values)

lower_bound = Q1 - 1.5 * IQR  // (Tukey rule value below that will be an Outlier)
upper_bound = Q3 + 1.5 * IQR
                // (Same as lower_bound but instead value above is outlier)

Outliers = df[[df["Math"] < lower_bound | df["Math"] > upper_bound)]
```

Key Important :

(i) For data cleaning as outliers can be mistakes

(ii) Anomaly detection. They may represent spoiled cases.

(iii) Hamsika.

Assignment 5 - Data Visulization of Distribution

Task: Plot histogram and boxplot of a data set. Interpret skewness and outliers.

Goal: Visulize frequency & dispersion

Libraries used: Pandas, Matplotlib.

Code Snippets:-

```
df. = pd. DataFrame ({"Age" : age}) //Store data in df

plt. hist(df["Age"], bins=8 ,color = "skyblue", edgecolor ="black")
plt. title ("Histogram of customer ages")
plt. xlabel ("Age")
plt. ylabel ("Frequency")
plt. show()

plt. boxplot (df["Age"], vert = False , patch_artist = True)
plt. title ("Boxplot of Customer Ages")
plt. xlabel ("Age")
plt. show ()
```

//We are using matplotlib to visulaize the data.
How we are using histogram to show the shape of the distribution of data (normal, skewed, uniform etc).

// Boxplots is used to summarize data with median, quartiles, spread and outliers.

Assignment 6 - Correlation Matrix (Intermediate)

Task: Load a dataset with multiple numerical columns. Compute correlation matrix and visualize with heatmap.

Goal: Practice coorelation in descriptive stats to see relationships.

Libs used : pandas, seaborn, matplotlib.

Code :

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt.

data = {
        "sales" : ....
        "Ads" : ....
        "Profit" : ....
        "weather" : ....
    }

df = pd.DataFrame (data)
print (df)

corr_matrix = df.corr()
print (corr_matrix)

plt.figure (figsize = (6,4))
sns.heatmap (corr_matrix, annot = True, cmap="coolwarm",
             linewidth = 0.5)
plt.title ("Coorrelation Matrix Heatmap")
plt.show ().
```

Profit

// Strong correlation -> Ads & Sales
// Weak " -> Weather v/s sales / Ads / Profits

Assignment 7 : Group-wise Descriptive Stats

Task : Group a dataset by category. Compute mean, median, variance for each group.

Goals : Learn group-level descriptive stats, often used in business insights and customer segmentation

Libraries used : Pandas

Code :

```
import pandas pd
data ={
    "Region" : [...  ],
    "Sales" : [... ],
    }
df = pd. DataFrame (data)
print (df)

grouped_ stats = df. groupby ("Region") [ "Sales" ].agg (
                Mean = "mean",
                Median = "median",
                Variance = "var"
               )
print (grouped_stats)

// The groupby() method divides the DataFrame into
groups based on the values in one or more
specified column.

// The agg() function then applies one or more agg. function
to each group.
// Finally the results are combined
```

[2]:
```python
import numpy as np

data = [12, 15, 20, 22, 25, 30, 18]

data_range = max(data)-min(data)

data_variance = np.var(data, ddof=1)

data_std = np.std(data, ddof = 1)


print("Range:", data_range)
print("Variance:", data_variance)
print("Standard Deviation:", data_std)
```

```
Range: 18
Variance: 36.904761904761905
Standard Deviation: 6.074928962939559
```

[2]:
```python
import pandas as pd

df = pd.read_csv("student_marks.csv")

print(df.head())

summary = df.describe()

print(summary)
```

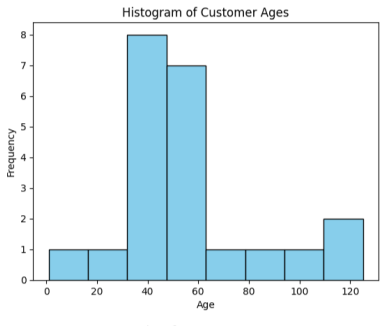
```
  Student  Math  Science  English
0       A    78       82       75
1       B    65       70       68
2       C    89       88       85
3       D    55       58       50
4       E    72       76       70
            Math    Science    English
count  10.000000  10.000000  10.000000
mean   73.200000  74.100000  70.600000
std    11.631375   9.926955  11.983322
min    55.000000  58.000000  50.000000
25%    65.750000  67.750000  65.750000
50%    73.000000  74.500000  71.000000
75%    80.250000  81.500000  77.250000
max    90.000000  88.000000  88.000000
```

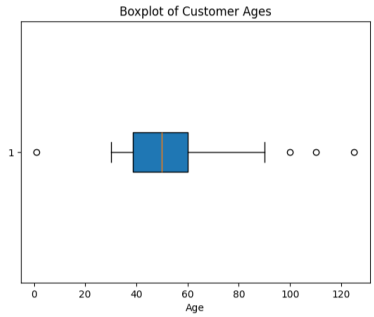
[5]:

```python
import pandas as pd

# Example dataset
data = {
    "Math": [78, 65, 89, 55, 72, 90, 68, 74, 81, 60, 5, 1000]  # notice the "5" is an outlier
}
df = pd.DataFrame(data)

# Step 1: Calculate Q1, Q3, IQR
Q1 = df["Math"].quantile(0.25)
Q3 = df["Math"].quantile(0.75)
IQR = Q3 - Q1

print("Q1:", Q1)
print("Q3:", Q3)
print("IQR:", IQR)

# Step 2: Define outlier bounds
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

print("Lower Bound:", lower_bound)
print("Upper Bound:", upper_bound)

# Step 3: Find outliers
outliers = df[(df["Math"] < lower_bound) | (df["Math"] > upper_bound)]
print("Outliers:")
print(outliers)
```

```
Q1: 63.75
Q3: 83.0
IQR: 19.25
Lower Bound: 34.875
Upper Bound: 111.875
Outliers:
     Math
10      5
11   1000
```

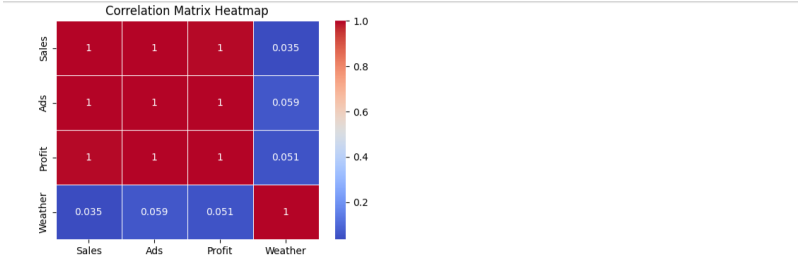
[10]:
```
import pandas as pd
import matplotlib.pyplot as plt

ages = [1,60, 68, 50, 100, 30, 32, 35, 36, 38, 110,
        41, 42, 45, 47, 50, 52, 54, 57, 60, 125, 90]

df = pd.DataFrame({"Age" : ages})

plt.hist(df["Age"], bins=8, color="skyblue", edgecolor="black")
plt.title("Histogram of Customer Ages")
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.show()

plt.boxplot(df["Age"], vert=False, patch_artist=True)
plt.title("Boxplot of Customer Ages")
plt.xlabel("Age")
plt.show()
```




Boxplot of Customer Ages
1
0
20
40
60
80
100
120
Age

[3]:
```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

data = {
    "Sales":  [200, 220, 250, 270, 300, 320, 340, 360, 400, 420],
    "Ads":    [20, 22, 25, 28, 30, 32, 34, 36, 40, 42],   # strongly related to Sales
    "Profit": [50, 55, 60, 70, 75, 80, 85, 90, 100, 110], # strongly related to Sales
    "Weather":[30, 25, 28, 32, 29, 27, 31, 26, 30, 28]    # almost unrelated to Sales/Ads/Profit
}

df = pd.DataFrame(data)
print(df)

corr_matrix = df.corr()
print(corr_matrix)

plt.figure(figsize=(6,4))
sns.heatmap(corr_matrix, annot = True, cmap="coolwarm", linewidth=0.5)
plt.title("Correlation Matrix Heatmap")
plt.show()
```

```
   Sales  Ads  Profit  Weather
0    200   20      50       30
1    220   22      55       25
2    250   25      60       28
3    270   28      70       32
4    300   30      75       29
5    320   32      80       27
6    340   34      85       31
7    360   36      90       26
8    400   40     100       30
9    420   42     110       28
             Sales       Ads    Profit   Weather
Sales    1.000000  0.999102  0.995973  0.035197
Ads      0.999102  1.000000  0.997000  0.058611
Profit   0.995973  0.997000  1.000000  0.051381
Weather  0.035197  0.058611  0.051381  1.000000
```


Correlation Matrix Heatmap
Sales
Ads
Profit
Weather
1
1
1
0.035
1
1
1
0.059
1
1
1
0.051
0.035
0.059
0.051
1
Sales
Ads
Profit
Weather
1.0
0.8
0.6
0.4
0.2

[4]:
```python
import pandas as pd

data = {
    "Region": ["North", "South", "East", "West", "North", "South", "East", "West", "North", "South"],
    "Sales":  [200, 180, 220, 210, 250, 190, 230, 205, 240, 195]
}

df = pd.DataFrame(data)
print(df)

grouped_stats = df.groupby("Region")["Sales"].agg(
    Mean = "mean",
    Median="median",
    Variance="var"
)
print(grouped_stats)
```

```
  Region  Sales
0  North    200
1  South    180
2   East    220
3   West    210
4  North    250
5  South    190
6   East    230
7   West    205
8  North    240
9  South    195
              Mean  Median    Variance
Region
East    225.000000   225.0   50.000000
North   230.000000   240.0  700.000000
South   188.333333   190.0   58.333333
West    207.500000   207.5   12.500000
```

[4]:
```python
import pandas as pd

data = {
    "Region": ["North", "South", "East", "West", "North", "South", "East", "West", "North", "South"],
    "Sales":  [200, 180, 220, 210, 250, 190, 230, 205, 240, 195]
}

df = pd.DataFrame(data)
print(df)

grouped_stats = df.groupby("Region")["Sales"].agg(
    Mean = "mean",
    Median="median",
    Variance="var"
)
print(grouped_stats)
```

```
  Region  Sales
0  North    200
1  South    180
2   East    220
3   West    210
4  North    250
5  South    190
6   East    230
7   West    205
8  North    240
9  South    195
              Mean  Median    Variance
Region
East    225.000000   225.0   50.000000
North   230.000000   240.0  700.000000
South   188.333333   190.0   58.333333
West    207.500000   207.5   12.500000
```